CHAPTER–IX

DIELECTRIC PROPERTIES OF SOLIDS

Dielectrics are basically electric insulators which ordinarily do not contain any free charge carriers for conduction. They, however, contain positive and negative charges which are bound together and hence could be affected by the applied electric fields. A brief description of the properties of dielectric solids in the presence of an external electric field is given in this chapter. These properties are important to understand the propagation of electromagnetic waves through the material media and fabricate various devices such as capacitors, microphones, etc.

9.1 POLARIZATION AND SUSCEPTIBILITY

When a dielectric is placed in an external electric field $\mathbf{E}_0$, the positive and negative charges are displaced from their equilibrium positions by very small distances (less than an atomic diameter) throughout the volume of the dielectric. This results in the formation of a large number of dipoles each having some dipole moment in the direction of the field. The material is said to be polarized with a *polarization* **P** defined as the dipole moment per unit volume of the material. As shown in Fig. 9.1, the effect of polarization is to reduce the magnitude of the external field $\mathbf{E}_0$. Thus the magnitude of the resultant field is less than the applied field, i.e., $E < E_0$. In vector notation, we may write

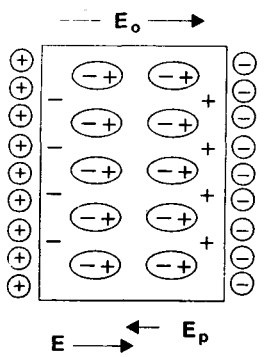


Fig. 9.1. *A dielectric slab placed in an electric field* $\mathbf{E}_0$ *produced by fixed charges (encircled) outside the slab. The internal polarization field* $\mathbf{E}_p$ *is assumed to be due to fictitious bound charges at the surface of the slab and is directed opposite to* $\mathbf{E}_0$.

$$\mathbf{E} = \mathbf{E}_0 + \mathbf{E}_p \tag{9.1}$$

The field $\mathbf{E}_p$ is called the *polarization field* as it tends to oppose the applied field E_0 within the material. For ordinary electric fields, the polarization **P** is proportional to the macroscopic field **E**. In SI units, it is expressed as

$$\mathbf{P} = \varepsilon_0 \chi_e \mathbf{E} \tag{9.2}$$

where ε_0 is the permittivity of free space and χ_e is the electric susceptibility. Thus, except for a

constant factor, the electric susceptibility is a measure of the polarization produced in the material per unit resultant electric field.

9.2 THE LOCAL FIELD

The electric field acting at the site of an atom or molecule is, in general, significantly different from the macroscopic electric field **E** and is called the *local field*. This field is responsible for polarization of each atom or molecule of a solid. For an atomic site with cubic symmetry, the local field is given by the *Lorentz relation,* i.e.,

$$\mathbf{E}_{loc} = \mathbf{E}_o + \mathbf{E}_p + \frac{\mathbf{P}}{3\varepsilon_o} = \mathbf{E} + \frac{\mathbf{P}}{3\varepsilon_0} \tag{9.3}$$

Thus, apart from the macroscopic field, the local field also contains a term which represents the field due to polarization of other atoms in the solid. The expression (9.3) gets modified with shape of the specimen.

9.3 DIELECTRIC CONSTANT AND POLARIZABILITY

The electric displacement vector for an isotropic or cubic medium can be defined as

$$\mathbf{D} = \varepsilon_o \varepsilon_r \mathbf{E} = \varepsilon_o \mathbf{E} + \mathbf{P} \tag{9.4}$$

where ε_r is called the *relative permittivity* or *dielectric constant* of the dielectric. It is a scalar quantity for an isotropic medium and is always dimensionless. The Eq. (9.4) can be used to define the dielectric constant as

$$\varepsilon_r = \frac{\varepsilon_o \mathbf{E} + \mathbf{P}}{\varepsilon_o \mathbf{E}} = 1 + \chi_e \tag{9.5}$$

Thus, like susceptibility, the dielectric constant is also a measure of the polarization of the material. Larger the polarization per unit resultant field, greater will be the dielectric constant of the dielectric.

The polarizability, α, of an atom is defined as the dipole moment per unit local electric field at the atom, i.e.,

$$p = \alpha E_{loc} \tag{9.6}$$

Thus polarizability is an atomic property whereas dielectric constant is a macroscopic property which depends upon the arrangement of atoms within the crystal. If all the atoms have the same polarizability and there are N number of atoms per unit volume, the polarization can be expressed as

$$P = Np = N\alpha E_{loc} \tag{9.7}$$

The general expression for polarization is, however, given by

$$P = \sum_j N_j \alpha_j E_{loc}(j) \tag{9.8}$$

where the summation is over all the atoms or atomic sites. In case of cubic symmetry, Eqs. (9.3) and (9.7) yield

$$P = N\alpha\left(E + \frac{P}{3\varepsilon_o}\right) \tag{9.9}$$

which, on rearranging the terms, gives

$$\chi_e = \frac{P}{\varepsilon_o E} = \frac{N\alpha/\varepsilon_o}{1 - \dfrac{N\alpha}{3\varepsilon_o}} \tag{9.10}$$

Using Eq. (9.5), we obtain

$$\varepsilon_r = 1 + \frac{N\alpha/\varepsilon_o}{1 - \dfrac{N\alpha}{3\varepsilon_o}}$$

or

$$\varepsilon_r = \frac{1 + \dfrac{2N\alpha}{3\varepsilon_o}}{1 - \dfrac{N\alpha}{3\varepsilon_o}} \tag{9.11}$$

This expression can be put in the form

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_o} \tag{9.12}$$

This is known as the *Clausius-Mossotti relation.* It relates the dielectric constant to the atomic polarizability provided the condition of cubic symmetry holds. In a more general form, it is expressed as

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{1}{3\varepsilon_o} \sum_j N_j \alpha_j \tag{9.13}$$

9.4 SOURCES OF POLARIZABILITY

The net polarizability of a dielectric material results mainly from the following three types of contributions :

(*i*) Electronic polarizability

(*ii*) Ionic polarizability

(*iii*) Dipolar or orientational polarizability

The extent to which a particular polarizability contributes depends on the nature of the dielectric and the frequency of the applied electric field.

9.4.1 Electronic Polarizability

The electronic polarizability arises due to the displacement of the electron cloud of an atom relative to its nucleus in the presence of an applied electric field as shown in Fig. 9.2. The polarization as well as the dielectric constant of a material at optical frequencies results mainly from the electronic polarizability. At optical frequencies, Eq. (9.13) may be written as

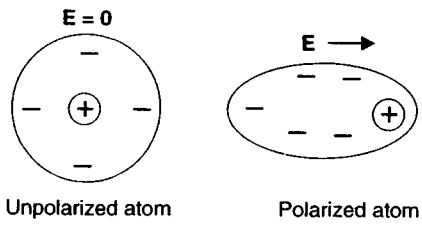


Fig. 9.2. *Electronic polarization in the presence of an external field.*

$$\frac{n^2-1}{n^2+2} = \frac{1}{3\varepsilon_0} \sum_j N_j \alpha_j \text{ (electronic)} \tag{9.14}$$

where ε_r has been replaced by n^2, n being the refractive index.

Classical Theory of Electronic Polarizability

An electron bound harmonically to an atom exhibits resonant absorption at a frequency given by

$$\omega_0 = \sqrt{\beta / m} \tag{9.15}$$

where β represents the force constant and m is the mass of the electron. If x is the displacement of the electron under the effect of the electric field E_{loc}, then we have

$$-eE_{loc} = \beta x = m\omega_0^2 x \tag{9.16}$$

The static electronic polarizability is calculated as

$$\alpha_e = \frac{p}{E_{loc}} = -\frac{ex}{E_{loc}} = \frac{e^2}{m\omega_0^2} \tag{9.17}$$

To obtain the frequency dependence of electronic polarizability, we treat the system as a simple harmonic oscillator. If ω is the frequency of the local field, the field at any time t is given by $E_{loc} \sin \omega t$ and the equation of motion may be written as

$$m\frac{d^2x}{dt^2} + m\omega_0^2 x = -eE_{loc} \sin \omega t \tag{9.18}$$

Substituting $x = x_0 \sin \omega t$, we obtain

$$m\,(-\omega^2+\omega_0^2)\,x_0 = -\,e\,E_{loc} \tag{9.19}$$

The amplitude of the dipole is given by

$$p_0 = -ex_0 = \frac{e^2 E_{loc}}{m\left(\omega_0^2 - \omega^2\right)}$$

which gives

$$\alpha_e = \frac{p_0}{E_{loc}} = \frac{e^2/m}{\omega_0^2 - \omega^2} \tag{9.20}$$

This expression gives the frequency dependence of electronic polarizability. The corresponding expression based on the quantum theory is given by :

$$\alpha_e = \frac{e^2}{m} \sum_j \frac{f_{ij}}{\omega_{ij}^2 - \omega^2} \tag{9.21}$$

where f_{ij} is known as the *oscillator strength* of the electric dipole transition between the atomic states i and j.

9.4.2 Ionic Polarizability

The ionic polarizability arises due to displacement of a charged ion relative to other ions in a solid. Assuming the forces near equilibrium as simple harmonic, the displacement Δx in the presence of an electric field E is given by

$$\beta \Delta x \cong eE \tag{9.22}$$

where β is the force constant. Thus the ionic polarizability is determined as

$$\alpha_i = \frac{p}{E} = \frac{e\Delta x}{E} \cong \frac{e^2}{\beta} \tag{9.23}$$

For $\beta \cong 20\ \mathrm{Nm^{-1}}$, $\alpha_i \cong 10^{-39}\ \mathrm{Fm^2}$. The ionic contribution is important at low frequencies. The sodium chloride, for example, has $\varepsilon_r \cong 5.6$ at low frequencies whereas the value reduces to about 2.25 at optical frequencies.

9.4.3 Dipolar Polarizability

A molecule, such as H_2O, having a permanent dipole moment is called a *dipolar* or *polar molecule* and a substance comprising such molecules is called a *dipolar substance.* The dipolar polarizability is the property of dipolar substances. In the absence of an external electric field, the dipoles have

random orientations and there is no net polarization. However, when the field is present, the dipoles orient themselves along the field and produce *orientational* or *dipolar polarizability*. The thermal agitation of the molecules tends to counteract the ordering effect of the electric field and an equilibrium state is reached wherein the different dipoles make all possible angles varying from zero to π radians with the field direction. The potential energy of such a molecule of dipole moment **p** oriented at an angle θ with the field direction (Fig. 9.3) is given by

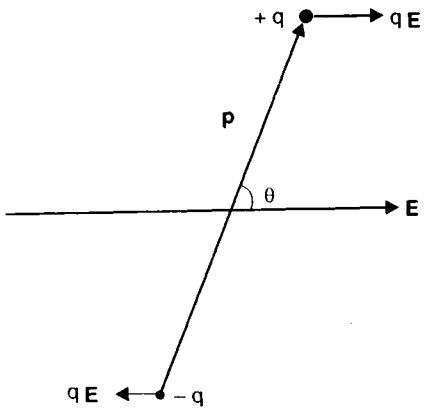


Fig. 9.3. *A molecule of dipole moment* **p** *placed in an electric field* **E**.

$$U = -\,\mathbf{p}.\mathbf{E} = -pE\cos\theta \tag{9.24}$$

According to the statistical mechanics, the number of dipoles having orientations between θ and $\theta + d\theta$, i.e., which lie within the solid angle $d\Omega$ or $2\pi\sin\theta d\theta$ is proportional to

$$\exp\left(-\frac{U}{kT}\right)d\Omega \quad \text{or} \quad \exp\left(\frac{pE\cos\theta}{kT}\right)2\pi\sin\theta d\theta \tag{9.25}$$

Since a dipole making an angle θ with the field direction contributes a component of dipole moment $p\cos\theta$ parallel to the field, the contribution of the above number of dipoles to the total polarization is

$$2\pi p\exp\left(\frac{pE\cos\theta}{kT}\right)\cos\theta\sin\theta\, d\theta \tag{9.26}$$

The average contribution to polarization is, therefore, given by

$$\bar{p} = \frac{P}{N} = \frac{\text{Total polarization due to all the dipoles}}{\text{Total number of dipoles}}$$

$$= \frac{p\int_0^\pi \exp\left(\frac{pE\cos\theta}{kT}\right)\cos\theta\sin\theta\, d\theta}{\int_0^\pi \exp\left(\frac{pE\cos\theta}{kT}\right)\sin\theta\, d\theta} \tag{9.27}$$

Putting

$$x = \frac{pE}{kT},\ y = \cos\theta \text{ and } dy = -\sin\theta\, d\theta,$$

Eq. (9.27) becomes

$$\bar{p} = \frac{p\int_{+1}^{-1} ye^{xy}dy}{\int_{+1}^{-1} e^{xy}dy}$$

or

$$\frac{\bar{p}}{p} = \frac{e^x + e^{-x}}{e^x - e^{-x}} - \frac{1}{x}$$

$$= \coth x - \frac{1}{x} \equiv L(x) \tag{9.28}$$

where $L(x)$ is called the *Langevin function*. The variation of $L(x)$ with x has been shown in Fig. 8.2. For $x << 1$ or $pE << kT$, i.e., for fields not too large and for temperatures not too low, we have

$$L(x) \cong \frac{x}{3} = \frac{pE}{3kT} \tag{9.29}$$

Using Eqs. (9.28) and (9.29), the polarization P becomes

$$P = N\bar{p} = \frac{Np^2E}{3kT} \tag{9.30}$$

The dipolar polarizability per molecule is, therefore, given by

$$\alpha_d = \frac{\bar{p}}{E} = \frac{p^2}{3kT} \tag{9.31}$$

Thus the existence of dipolar polarizability depends on whether the molecules possess a permanent dipole moment. The dipolar polarizability also depends on temperature in accordance with Eq. (9.31). At room temperature, $\alpha_d \sim 10^{-39}$ Fm^2 which is comparable to electronic polarizability.

9.5 FREQUENCY DEPENDENCE OF TOTAL POLARIZABILITY

The total polarizability of a dielectric is given by the expression :

$$\alpha = \alpha_e + \alpha_i + \alpha_d \tag{9.32}$$

It decreases with increase in frequency as shown in Fig. 9.4. This type of